SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	27 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Flask-Login integration for secure user login. Authorized personnel must provide verified organizational emails for administrative access.	High
2	Authorization levels	Role-Based Access Control (RBAC). Roles are Meteorologist, Coordinator, Field Responder, and Admin. Defines specific read/write permissions for map data, metrics input, and alert triggers for each role.	High
3	External interfaces	Continuous real-time telemetry stream from external weather APIs (OpenWeatherMap). Secure outbound REST webhook integration with the Twilio API gateway for SMS broadcasting.	High
4	Transactions processing	Immediate formatting of weather input vectors into DataFrames, processed by the pre-trained XGBoost matrix pipeline. Returns risk results within milliseconds.	High
5	Reporting	Leaflet.js interactive map with risk polygons. Download utility to export predictive analytics as CSV summaries.	Medium
6	Business rules, etc.	Trigger threshold override logic. If flood probability > 75%, system flags district red and pre-populates alert payload text.	High
7	Compliance to laws or regulations	Localized environmental data sharing standards alignment. Digital communication privacy terms compliance compliance for SMS broadcasts.	Medium
8	Other	Manual data override for authorized meteorologists to test simulation models or correct system alerts during sensor outages.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The model must provide risk classification scores quickly upon data ingestion or refresh.	XGBoost model execution and page load times must complete in < < 1.5 seconds.
2	Scalability	The infrastructure must auto-scale to handle sudden traffic spikes during major storm transitions.	Cloud infrastructure must must support up to 5,000 concurrent active map users without degradation.
3	Security & Data Privacy	All API communications, administrative sessions, and logging databases must be fully protected.	Implementation of HTTPS TLS 1.3 and AES-256 bit encryption on stored telemetry histories.
4	Reliability & Availability	The service must remain operational and accessible during peak disaster management windows.	The service must guarantee a 99.95% system uptime across all instances per severe weather season.
5	Usability & Accessibility	The mapping dashboard must be simple to navigate under high-stress crisis scenarios with clear visibility.	High-contrast UI elements complying fully with WCAG 2.1 AA design guidelines.
6	Maintainability & Portability	System updates, algorithm changes, and cloud migrations should be completed without significant disruption.	Entire application encapsulated inside Docker containers for seamless migration across IBM Cloud clusters.